



Faculty of Sciences
Faculty of Mathematics, Statistics, and Computer Science
Statistical Machine Learning — Homework 3

Due Date: Tue, 25 Nov 2025 (23:59)

Introduction.

In this homework, we will explore applying **Principal Component Analysis (PCA)** and two classical classifiers—**K-Nearest Neighbors (KNN)** and **Logistic Regression**—to a binary image-classification problem. The goal is to classify pictures of **Manchester United** players versus **Chelsea** players.

In this assignment, you will:

- Perform PCA to reduce the dimensionality of image data and interpret principal components.
 - Apply **KNN** as a non-parametric, nonlinear classifier and evaluate its performance under different choices of k .
 - Apply **Logistic Regression** as a linear classifier and study the effect of **L2 and L1 regularization** on classification performance.
 - Compare linear vs. nonlinear decision boundaries and analyze when each method succeeds or fails.
 - Evaluate model performance using appropriate metrics and discuss the trade-offs between dimensionality reduction and predictive accuracy.
-

Part 0. Background & Dataset.

In the legendary **2008 UEFA Champions League Final**, Manchester United once again demonstrated their superiority on the biggest stage by defeating Chelsea—reminding the football world that some clubs lift trophies, while others merely slip on them. Chelsea were sent home empty-handed, forced to wait four more years before finally stumbling their way to their first Champions League title. Meanwhile, the **best player of all time, Cristiano Ronaldo**, lifted his *first* Champions League trophy—because greatness starts early, unlike Chelsea’s European history.

For this homework, we will use images taken from that iconic match (including Cristiano Ronaldo graciously kissing the trophy) as well as a few other matches involving both clubs. The dataset also contains the unforgettable moment where **John Terry**, Chelsea’s captain, produced one of the most dramatic slip-and-cry combinations in football history. His tearful reaction after missing the decisive penalty has become a timeless masterpiece, studied in memes and now in this homework. These images form our dataset for PCA and classification.



(a) Cristiano Ronaldo celebrating with the trophy.



(b) John Terry moments after turning the Champions League final into a slip-and-slide tutorial.

Part 1. Feature Extraction Using PCA.

Now that our dataset is prepared, we proceed to extract meaningful features from the images. Since raw pixel representations are extremely high-dimensional, we apply **Principal Component Analysis (PCA)** to obtain a lower-dimensional set of principal components that capture the most important sources of variation in the data.

- Apply PCA to the full data set of images and compute the principal components.
- Using the techniques introduced in the tutorial session (e.g., scree plot, cumulative explained variance, elbow method), determine an appropriate number of components to retain.
- Clearly report the number of principal components you selected and justify your choice based on the diagnostic plots.
- Reconstruct several images using:
 - a moderate number of components,
 - your selected number of components,
 - and a large number of components.
- Compare each reconstructed image to its original version and briefly explain how the number of components affects reconstruction quality.

Part 2. Classification and Cross-Validation.

After extracting PCA features, we now evaluate several classification models using an 80/20 train–test split and cross-validation procedures. Our goal is to compare linear and nonlinear models and study the effect of regularization in logistic regression.

- Split the dataset into **80% training** and **20% testing** (set a fixed random seed).
- On the training set, perform **k -fold cross-validation** to evaluate the performance of the following models:
 - **K-Nearest Neighbors (KNN)** classifier,
 - **Logistic Regression** (no regularization),
 - **Logistic Regression with L1 regularization**,

- **Logistic Regression with L2 regularization.**
 - For the regularized logistic regression models:
 - Use a separate **validation set** (or nested CV) to tune hyperparameters such as the regularization strength λ (or inverse parameter C).
 - Report the best hyperparameter values found for both L1 and L2 models.
 - For each trained classifier, compute the following metrics on the **held-out 20% test set**:
 - Precision,
 - Recall,
 - F1 score,
 - Accuracy,
 - Confusion matrix.
 - Create a comparison table summarizing the performance of all four models based on the above metrics.
-

Part 3. Nonlinear Feature Engineering.

To further improve predictive performance, we now expand the feature space by introducing **nonlinear transformations** of the PCA features. These transformations allow linear models—especially logistic regression—to capture more complex structures in the data.

- Construct additional nonlinear features using transformations such as:
 - **Polynomial features** (e.g., degree 2 or 3),
 - **Logarithmic transforms** (applied only to components with strictly positive values or make them strictly positive),
 - **Interaction terms** between selected principal components.
- Combine the original PCA features with the newly created nonlinear features to form an expanded feature set.
- Retrain all classifiers from Part 2 using this enriched feature space:
 - KNN,
 - Logistic Regression,

- Logistic Regression with L1 regularization,
- Logistic Regression with L2 regularization.
- Repeat the complete evaluation pipeline from the previous section:
 - Perform cross-validation on the training set,
 - Use a validation set to tune L1 and L2 regularization strengths,
 - Compute test-set performance metrics:
 - * Precision,
 - * Recall,
 - * F1 score,
 - * Accuracy,
 - * Confusion matrix.
- Report all results in a structured comparison table, clearly indicating whether nonlinear transformations improved the predictive metrics relative to the baseline PCA-only models.

Part 4. Final Model Selection & Error Analysis.

In this final stage, you will identify the best-performing model from all previous sections and conduct a deeper analysis of its classification errors.

- Based on the precision, recall, F1 score, accuracy, and confusion matrices from earlier sections, select the **best overall model**. Clearly justify your choice.
- Using the chosen model, extract the instances from the test set that were:
 - **False Positives (FP)**: images incorrectly predicted as Manchester United,
 - **False Negatives (FN)**: images incorrectly predicted as Chelsea.
- Display several representative FP and FN images in your report. For each example:
 - Show the predicted label and the true label,
 - Briefly describe any visual patterns that might explain the model’s mistake.
- Provide a short discussion explaining *why* you think the model failed to correctly classify these specific instances. Possible reasons include:
 - the limited number of PCA components reducing discriminative information,

- nonlinear structures not fully captured even after transformations,
 - similar-looking jerseys, poses, or lighting conditions,
 - noisy or low-resolution input images,
 - insufficient training examples for certain visual patterns.
- Conclude by summarizing what these misclassifications reveal about the limitations of your best model and how more advanced methods could further reduce such errors.

Submission Checklist (no points). Compile to PDF, include your name/ID, label sub-parts, and attach any plots. If you used AI tools or external resources, briefly disclose: which tool/source and how it was used.

Problem	1	2	3	4	Total
Points	/25	/25	/25	/25	/100